Silent Voice Rig: Fused EMG Model Plan

Oct 23, 2026 · @Austin Wang

Mouthed speech becomes on-screen text and a cloned voice in real time. Mouth open: camera and EMG both propose sentences, and EMG, camera and a local LLM judge them together. Mouth closed: EMG alone picks one of ~10 pitch lines. The novelty is speed.

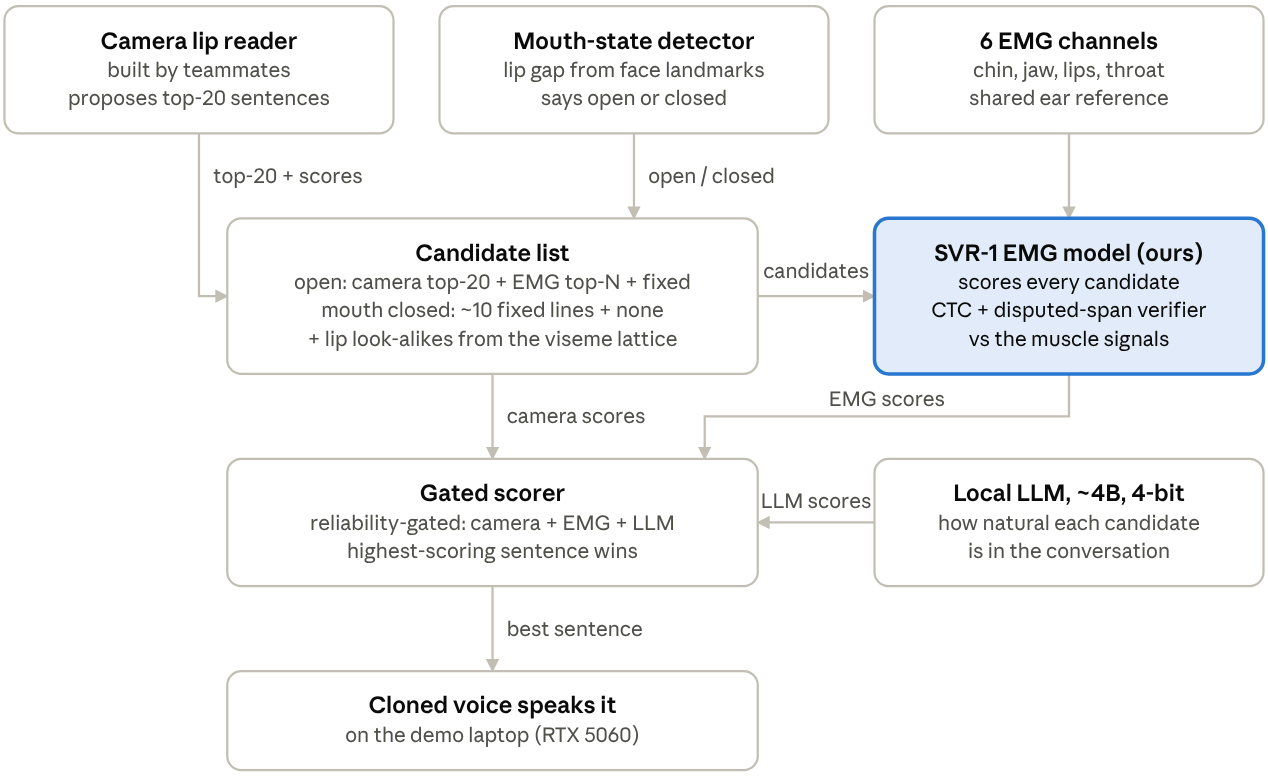
Status

Baselines are trained; SVR-1 is written and reviewed but not trained yet, because the H100 jobs ran before its code landed. Nothing is lost: local code and the bucket are identical.

Part	State	Next step
Gaddy baselines (H100/H200)	Done. 6-channel retrain G6: 33.2% WER. 8-channel retrain: 29.4% vs Gaddy's published 27.1%	G6 is the bar SVR-1 must beat
SVR-1 code	Done; reviewed twice; 3 crash bugs and 2 big speed losses fixed; all CPU unit checks pass	Cluster: articulator targets, smoke test, then the 11-run sweep
SVR-1 training	Not started	`smoke_svr1.sh SVR1_full`, then `svr1_sweep.sh`
Camera baselines (Auto-AVSR, +LLM, BRAVEn, LipLearner)	Code done; cluster smoke fixes merged	Run on the wearer's recordings
Wearer fine-tuning + benchmarks	`svr/finetune.py` done, tested on synthetic data	Needs the day-2 recordings
Fusion gate + real-time loop	Designed (figure panel G)	Build on day 3
Code sync	Local = bucket, with the cluster's fixes merged	Pull before every cluster session

How it works

When the mouth closes, a learned gate shifts trust from camera to EMG



fused system · 3 inputs, one scorer

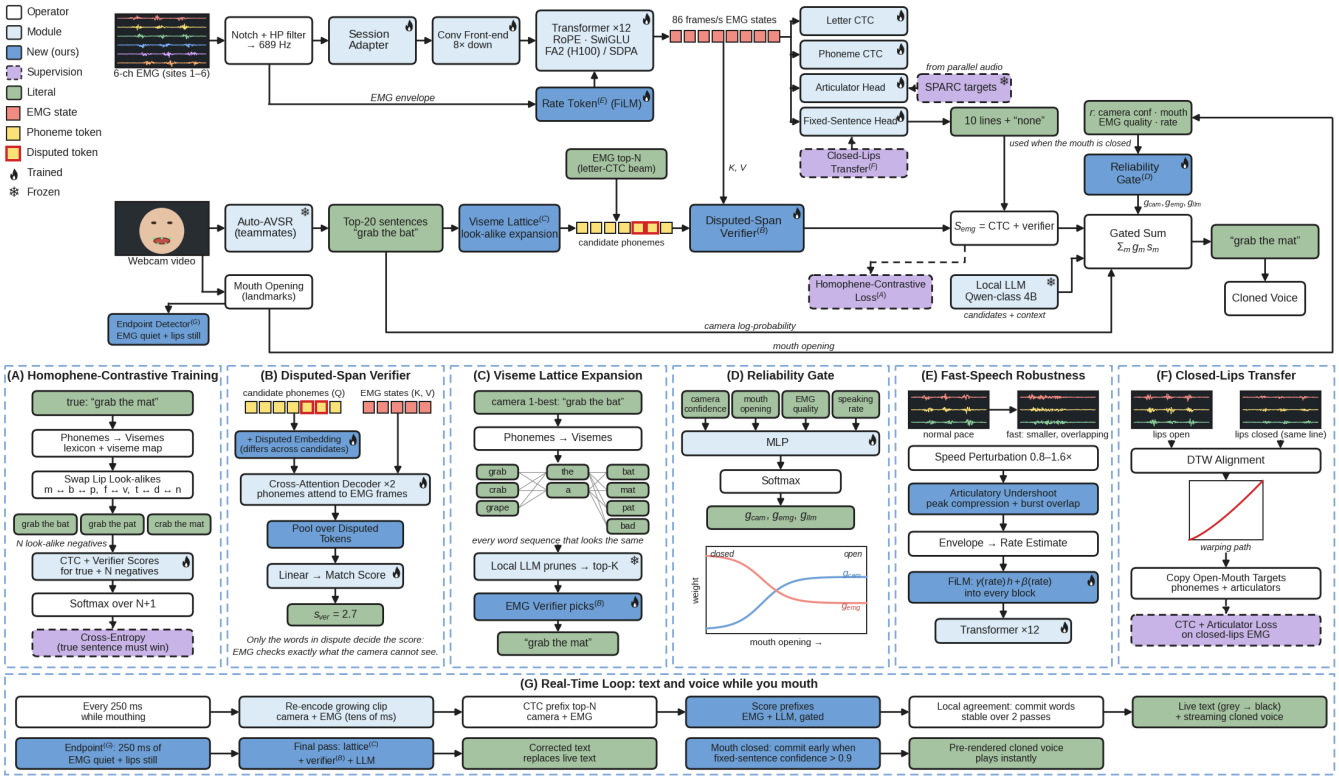
Both streams propose candidate sentences and both are judged; closing the mouth shrinks the candidates to the pitch lines and hands the decision to EMG.

	Mouth open	Mouth closed
Inputs	Camera + 6 EMG channels	6 EMG channels
Who proposes	Camera top-20, EMG top-N (its own letter decoder), lip look-alike expansion	The ~10 pitch lines + "none"
Who judges	EMG score + camera score + local LLM, weighted by the reliability gate	EMG alone
Target	Low WER at 150–180+ words per minute	Near-perfect pick; pre-rendered voice starts at once

- No mode switch: the gate slides trust from camera to EMG as the mouth closes.
- Electrodes: Gaddy sites 1–6 (two chin sites, two cheeks, jaw, throat), reference below the left ear. Sites 7–8 dropped: they rank last and sit on the upper cheek.
- Ownership: we train the EMG model and build the fusion; teammates own hardware, recorder, camera and the cloned voice (Cartesia).

SVR-1, the EMG model

SVR-1 (45M parameters) turns 6-channel EMG into 86 frames per second, writes its own guesses with a letter decoder, and scores any candidate sentence. A separate doc, SVR-1 explained, walks through every box of the figure and the real-time levers.



Dark blue = our modules; flame = trained, snowflake = frozen; panels A–F = contributions; panel G = the real-time loop.

Contributions

Each one is a switch in the code, so the sweep measures what it adds.

#	Contribution	What it does
A	Homophene-contrastive training	Teaches EMG to pick the true sentence over lip look-alikes (bat / mat / pat)
B	Disputed-span verifier	Checks only the words the candidates disagree on against the muscle signal
C	Viseme-lattice expansion	Reaches sentences outside the camera's top-20 that look the same on the lips
D	Reliability gate	Per-sentence trust in camera, EMG and LLM from mouth opening, confidence, signal quality and pace
E	Fast-speech robustness	Speed and articulatory-undershoot augmentation, plus a speaking-rate token
F	Closed-lips transfer	Open-mouth repetitions supervise closed-mouth EMG through time alignment (DTW)
G	Real-time loop	Live text every 250 ms; a word commits once two passes agree

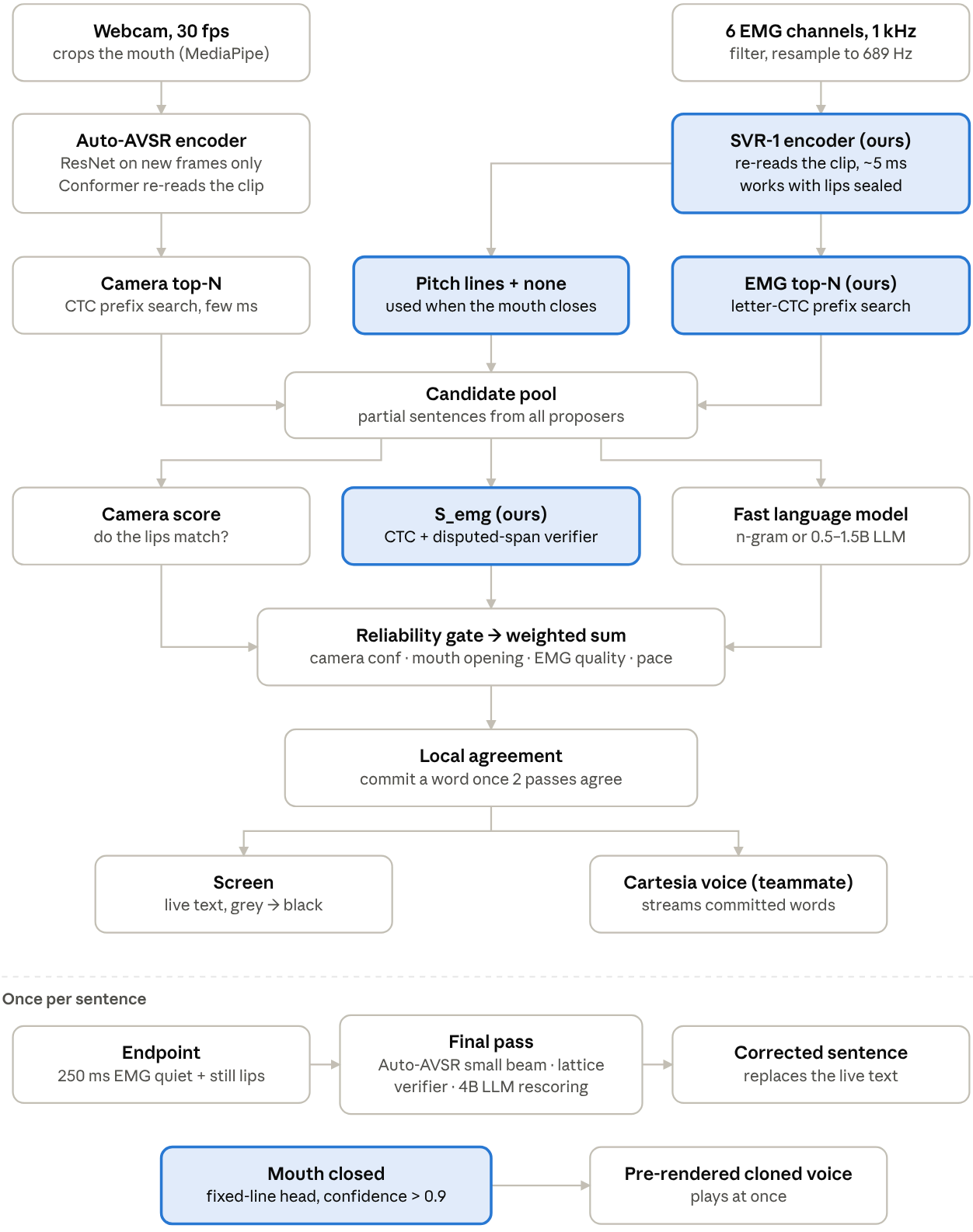
Built on and credited: articulator targets from audio (SPARC, Lee et al. 2025), 8-to-6-channel pretraining (Hwang et al. 2026), letter + phoneme CTC (Gaddy & Klein 2021).

Real-time pipeline

Target: each word on screen ~0.3–0.5 s after it is mouthed, the cloned voice streaming right behind it. Auto-AVSR is an offline model (it reads the whole clip), but we stream it by re-reading the growing clip every 250 ms. That needs no retraining, and it beats the 0.8 s delay of PyTorch's streaming model.

Every 250 ms both streams propose, three judges score, stable words go out

Live loop on the laptop (RTX 5060); highlighted boxes are SVR-1, our EMG model



end-to-end real-time pipeline · live loop + once-per-sentence pass

SVR-1 is the highlighted part; everything else is the camera, fusion, language model and voice around it.

1. Every 250 ms: run the camera front end on the new frames only, then re-encode the clip so far (camera + EMG).
2. Both streams propose top-N partial sentences with a fast CTC prefix search; no slow attention beam search while mouthing.
3. Score the candidates with EMG + camera + a fast language model, through the gate.
4. Commit a word once two consecutive passes agree; it turns from grey to black and streams to Cartesia.
5. Sentence end (250 ms of EMG quiet + still lips): the final pass adds look-alike expansion, the verifier and the 4B LLM, and corrects the text. Mouth closed: a pitch line commits at confidence > 0.9 and its pre-rendered voice plays at once.

Step	Budget (estimate, measured on the laptop day 3)
Camera front end, new frames	~1–2 ms per frame
Camera encoder re-read, 3–5 s clip	~10–30 ms
SVR-1 re-read	~5 ms
CTC prefix search, both streams	~2–5 ms
Live language scoring (n-gram, or a 0.5–1.5B LLM)	<1 ms / ~20–40 ms
Whole tick	well under the 250 ms tick
Final pass at sentence end (4B LLM, verifier)	~150–250 ms

The voice (Cartesia, teammate) is the largest remaining delay; our side only has to deliver stable words early.

Results so far

On our 6 sites, Gaddy's architecture retrained (G6) reaches 33.2% WER, but its look-alike accuracy falls from 84% to 49% at 1.5× speed: that drop is what SVR-1 targets. Berkeley test set, 99 sentences.

Model	Channels	WER, beam + LM (%)	Homophene-10 top-1 (%)	Same at 1.5× speed (%)
Gaddy's released model	8	27.1	88.3	–
Our retrain, Gaddy recipe	8	29.4	89.4	68.1
Gaddy's model, 2 channels zeroed	6	64.4	55.3	38.3
G6: retrained on 6 channels (bar to beat)	6	**33.2**	**84.0**	**48.9**
SVR-1	6	pending	pending	pending

- Forward pass: 3.9 ms per 3 s clip on an H200.
- The first seed of the 8-channel retrain collapsed (CTC stuck on blanks for 43 epochs); seed 1 trained normally. Watch dev WER early in every run.

Benchmarks

Berkeley benchmarks pick the best model now; the wearer's held-out recordings decide the demo. Headline chart: WER vs words per minute, camera alone vs fused.

Benchmark	Question it answers	Metric
Homophene judge (10 / 50 candidates)	Can EMG pick the true sentence among lip look-alikes?	Top-1
Fast-speech judge (1.25× / 1.5×)	Same, at speed	Top-1
Open vocabulary	How good is the encoder alone?	WER, CER
Wearer set E, three paces	Ours vs Auto-AVSR, Auto-AVSR + LLM, BRAVEn	WER vs WPM
Wearer fixed lines	Mouth closed: SVR-1 vs LipLearner (5-shot)	Top-1, false triggers on rest

Benchmark	Question it answers	Metric
Camera n-best + EMG	First fused number, before the gate exists	WER
Laptop latency	Real time?	Mouthed → shown, mouthed → heard

Data to record on the wearer

About 2.5 hours over two sessions on day 2, with camera and EMG on one clock. Format: `svr/recordings.py`.

Set	What	Amount	Use
A	The ~10 mouth-closed pitch lines	50 closed + 10 open each	Fixed-line path; closed-lips transfer
B	Rest, swallowing, random movement, talking	~100 clips	The "none" class
C	Varied sentences at ~100, ~150, 180+ WPM	~400	Fine-tuning; camera mistakes as training negatives
D	Open-mouth pitch lines	10 each	Demo robustness
E	Never-trained sentences, three paces	~100	The honest test numbers
F	Read aloud with a microphone	~100	Wearer's own articulator targets

- Mouth closed = lips sealed, tongue and jaw still actively articulating. A still face decodes at chance.
- Electrodes come off between sessions, so next-session accuracy is real; mark the skin.

Next steps

Day 2: train SVR-1 while the recordings happen. Day 3: build the fusion and real-time loop on the laptop.

Day 2

- [ ] Cluster: SPARC articulator targets, then `bash cluster/smoke_svr1.sh SVR1_full` until it prints SMOKE PASS
- [ ] Launch `SVR1_full`, then the rest of `cluster/svr1_sweep.sh` (8-to-6, six ablations, two sizes)
- [ ] Record sets A–F; run `svr.make_g2p` on the new prompts
- [ ] Run the camera baselines on the recordings (writes the n-best SVR-1 trains against)
- [ ] Fine-tune the sweep winner and G6 on the wearer: `svr.finetune`

Day 3

- [ ] Streaming camera: per-frame front end, re-read every 250 ms, CTC prefix search
- [ ] Fusion gate + look-alike expansion, trained on set C
- [ ] Local agreement, endpointing, Cartesia hand-off; measure latency on the RTX 5060
- [ ] WER-vs-WPM chart; rehearsal; fixed-line-only fallback

Risks

Risk	Mitigation
Closed-mouth EMG is weak if the wearer keeps a still face	Rehearse active tongue and jaw behind sealed lips; 50 reps per line; closed-lips transfer
Electrodes shift before the demo	Refit the session adapter from ~1 min of calibration; mark the skin
A training run collapses (seen once with the Gaddy recipe)	Watch dev WER in the first epochs; rerun with another seed
Camera's last word or two are unreliable on partial clips	Commit only words that two passes agree on; the final pass corrects
EMG adds little at speed	The gate falls back to camera + LLM; show the measured chart either way
Laptop runs out of its 8 GB	Smaller LLM for the live pass; 4B only at sentence end

Open: confirm sites 1–6 with hardware; recording time; the camera's n-best output format.